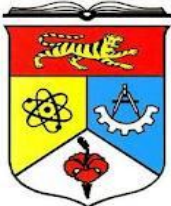


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ASASIpintar Programme (Pre-University level)

1. **Course Code:** PNAP0123

2. **Course Name:** Physics I

3. **Credit Unit:** 3

4. **Course Status:** Core

5. **Synopsis:**

This course provides a systematic introduction to the main principles of physics and emphasizes the development of conceptual understanding and problems solving. The course provides instruction in each of the following areas:

Physical Quantities & Measurement, Kinematics in 1-D, Kinematics in 2-D: Projectile Motion, Forces and Dynamics, Uniform Circular Motion, Work and Energy, Linear Momentum, Rotational Motion, Static Equilibrium and Centre of Mass, Vibration and Waves, Fluids

6. **Pre-requisite (if any):** SPM Physics

7. **Course requirements¹ to sit for examination:**

(Refer to *Buku Panduan Program ASASIpintar UKM* under Program Regulation: 7. Other Provisions, 7.1)

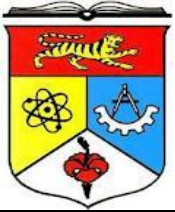
A student will not be permitted to sit for an examination of a course if the director has certified that the candidate has not fulfilled at least 70% of the course's requirement as set by Pusat GENIUS@Pintar Negara.

(Rujuk kepada *Buku Panduan Program ASASIpintar UKM* sesi akademik semasa di bawah Peraturan Program: 7. Peruntukan-peruntukan Lain, 7.1)

Seseorang pelajar tidak akan dibenarkan mengambil peperiksaan sesuatu kursus sekiranya Pengarah telah memperakui calon berkenaan telah tidak mengikuti sekurang-kurangnya 70% daripada keperluan kursus tersebut seperti yang ditetapkan oleh Pusat GENIUS@Pintar Negara.

Keperluan Kursus merupakan komponen penilaian berterusan yang boleh meliputi peratusan kehadiran, bilangan/peruntukan markah tugas/laporan yang telah dihantar, bilangan/pruntukan markah ujian yang telah diduduki serta komponen pentaksiran lain sepanjang minggu pengkuliah. Pelajar yang dihalang untuk menduduki peperiksaan perlu dikenal pasti selewat-lewatnya dua minggu sebelum peperiksaan bermula dan tertakluk kepada kelulusan Timbalan Pengarah, Pusat Pengurusan Akademik.

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8. References:

1. Serway, R.A., Vuille, C. & Hughes, J. *College Physics*, 12th edition 2025, Cengage Learning. (this course's textbook)
2. Giancoli, D.C., *Physics: Principles with Applications*, 7th edition 2016, Pearson Education International.
3. Cutnell, J.D. & Johnson, K.W., *Physics*, 12th edition 2022, John Wiley & Sons Singapore Pte. Ltd.
4. Hewitt, P.G., *Conceptual Physics*, 13th edition 2023, Pearson Education.

9. Course Learning Outcomes (CLO):

CLO 1: Students will demonstrate the ability to think critically and to use appropriate concepts to analyze qualitatively problems or situations involving the fundamental principles of physics.

Pelajar akan mempamerkan kebolehan untuk berfikir secara kritis, dan menggunakan konsep-konsep yang bertepatan bagi menganalisa secara kualitatif masalah atau situasi berdasarkan prinsip-prinsip asas fizik.

CLO 2: Students will demonstrate the ability to use appropriate mathematical techniques and concepts to obtain quantitative solutions to problems in physics.

Pelajar akan mempamerkan kebolehan menggunakan teknik dan konsep matematik yang bersesuaian untuk memperoleh penyelesaian kuantitatif terhadap masalah-masalah fizik.

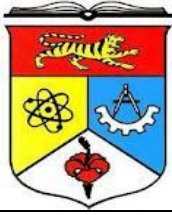
CLO 3: Students will demonstrate basic experimental skills by the practice of setting up and conducting an experiment with due regards to minimizing measurement error.

Pelajar akan mempamerkan kemahiran amali asas menerusi pemasangan peralatan dan pelaksanaan ujikaji dengan mengambil kira keperluan meminimumkan ralat pengukuran.

CLO 4: Students will demonstrate basic communication skills by working in groups on laboratory experiments through thoughtful discussion and interpretation of data.

Pelajar akan mempamerkan kemahiran berkomunikasi semasa pelaksanaan ujikaji secara berkumpulan menerusi perbincangan bernas dan penafsiran data pengukuran.

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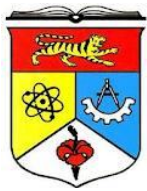
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Course Information

Assessment Methods:	continuous assessment	Topical Tests	15
		Physics Lab Report	15
		Online Assignment: Cengage WebAssign Interactive	10
		Mid Semester Examination	20
		Final Examination	40
	Total		100
Teaching & Learning Method:	Lecture, tutorial, laboratory experiments, independent learning, cooperative learning, small group discussion and problem-based learning		
Instructors (room no.) Tel / e-mail:	1. Dr. Nor Azah Nik Jaafar (Room no. 21) 019-2631409 / norazah_nj@ukm.edu.my 2. Dr. Nor Farhah Razak (Room no. 13) 89217570 / nfarhah@ukm.edu.my 3. Dr. Nurul Izzah Mukri 013-2509857 / nurulizzahmukri@gmail.com 4. Mr. Afiq Dzuan b. Mohd. Azhar 013-9709230 / afiqdzuan@ukm.edu.my		
Contact hours per week:	▪ 4 hours lecture per week ▪ 1-2 hours tutorial or 2 hours laboratory experiment per week		
Venue:	✚ This course will be administered via UKM Folio ✚ Teaching and Learning (TnL) face to face (f2f) session will be conducted at the following venues: i. Auditorium PPPN ii. Physics Lab 1 iii. Physics Lab 2 ✚ TnL online session will be conducted in any of these platforms: MS Teams, Zoom, Google Meet		
Important Dates:	Mid semester Examination: 24 - 30 August 2026 Final semester Examination: 16 - 22 November 2026		

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10. Teaching Plan:

Week	Topics	ML	T	L	PBL	SGD	CL	EX	IL	Total learning hours
1	Chapter 1: Introduction, Measurement and Estimating	2	1	-	-	-		0.5	1	4.5
2	Chapter 2: Kinematics in One Dimension (Part 1, 2)	4	1	-	1	-	-	0.5	1	7.5
3	Chapter 3: Kinematics in Two Dimension (Part 1, 2, 3)	5	-	2	1	-	-	0.5	1	9.5
4	Chapter 4: Dynamics and Newton's Laws of Motion (Part 1, 2)	3	1	-	1	-	-	0.5	1	6.5
5	Chapter 4: Dynamics and Newton's Laws of Motion (Part 3, 4)	4	1	-	1	-	-	-	-	6
6	Chapter 5: Circular Motion and Gravitation (Part 1, 2)	4	1	-	1	-	-	0.5	1	7.5
7	Chapter 6: Work and Energy (Part 1, 2)	4	-	2	1	-	-	0.5	1	8.5
8	STUDY LEAVE & REVISION	4	1	-	-	-	-	-	-	5

MID SEMESTER EXAMINATION WEEK		-	-	-	-	-	-	2	4	6
9	Chapter 7: Linear Momentum (Part 1, 2)	4	1	-	1	-	-	0.5	1	7.5
10	Chapter 8: Rotational Motion (Part 1, 2, 3)	6	1	-	1	-	-	0.5	1	9.5
11	Chapter 9: Static Equilibrium; Elasticity and Fracture (Part 1, 2)	4	1	-	1	-	-	0.5	1	7.5
12	Chapter 10: Oscillations and Waves (Part 1, 2)	4	1	-	1	-	-	0.5	1	7.5
13	Chapter 10: Oscillations and Waves (Part 3, 4)	4	-	-	1	-	-	-	-	5
14	Chapter 11: Sound (Part 1, 2)	4	1	2	1	-	-	-	-	8
15	Chapter 12: Fluids (Part 1, 2)	4	1	-	1	-	-	-	-	6
16	Chapter 12: Fluids (Part 3)	6	-	-	1	-	-	-	-	7
17	DISCUSSION: Problems Solvings Chapter 7-12	2	-	-	-	-	-	-	-	2
Total Learning Hours / Semester		68	12	6	14	0	0	7	14	121
Required Total Notional Credit Hours									120	
Credit Hours									3.0	

No.	Teaching Methodology	Acronym
1.	Mass Lecture	ML
2.	Tutorial	T
3.	Lab	L
4.	Inquiry Based Learning (IBL)	IBL
5.	Problem Based Learning (PBL)	PBL

No.	Teaching Methodology	Acronym
6.	Small Group Discussion (SGD)	SGD
7.	Cooperative Learning	CL
8.	Research Project	RP
9.	Independent learning	IL
10.	Examination	EX

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11. Assessment Plan:

No.	Course Learning Outcomes (CLO)	Bloom' s Taxonomy	PLO	Method of Delivery	Assessment Method		
					Cont. Assmt.	Mid Exam	Final Exam
1.	Students will demonstrate the ability to think critically and to use appropriate concepts to analyze qualitatively problems or situations involving the fundamental principles of physics.	C4	PLO 1	Lecture, Tutorial	40	20	40
2.	Students will demonstrate the ability to use appropriate mathematical techniques and concepts to obtain quantitative solutions to problems in physics.	C4	PLO 1	Lecture, Tutorial			
3.	Students will demonstrate basic experimental skills by the practice of setting up and conducting an experiment with due regards to minimizing measurement error.	P4	PLO 3,	Lab experiment			
4.	Students will demonstrate basic communication skills by working in groups on laboratory experiments through thoughtful discussion and interpretation of data.	A4	PLO 4	Lab experiment			
Total (%)					40	20	40

PROGRAM LEARNING OUTCOMES (PLO)

Graduates of ASASIpintar program will be able to:

- PLO 1** master an understanding of facts, concepts, principles and processes in various fields and apply it in everyday life
menguasai kefahaman dalam fakta, konsep, prinsip dan proses dalam pelbagai bidang dan mengaplikasikannya dalam kehidupan seharian
- PLO 2** practise high moral values through ethical behavior, show positive and responsible attitude to oneself, others and the environment
mengamalkan nilai-nilai moral yang tinggi melalui tingkah laku yang beretika, bersikap positif dan bertanggungjawab kepada diri sendiri, orang lain dan persekitaran
- PLO 3** show competency in searching, managing and integrating information for lifelong learning
menunjukkan kecekapan mencari, mengurus dan mengintegrasikan maklumat bagi amalan pembelajaran sepanjang hayat
- PLO 4** exhibit strong leadership skills, able to communicate effectively and work in a team
mempamerkan kemahiran kepimpinan yang tinggi, mampu berkomunikasi secara berkesan dan boleh bekerja dalam pasukan
- PLO 5** articulate critical, creative and innovative thinking, while able to solve problems intelligently in various situations and able to use high-technology applications
mengartikulasikan pemikiran secara kritis, kreatif dan inovatif, serta bijak menyelesaikan masalah dalam pelbagai situasi dan berkeupayaan menggunakan teknologi tahap tinggi

TAXONOMY CODES

Cognitive Domain

C1 Knowledge
C2 Comprehension
C3 Application
C4 Analysis
C5 Synthesis
C6 Creating

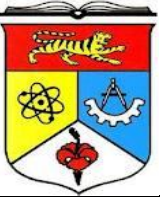
Psychomotor Domain

P1 Perception
P2 Set
P3 Guided Response
P4 Mechanism
P5 Complex Overt Response
P6 Adaptation
P7 Origination

Affective Domain

A1 Receiving Phenomena
A2 Responding to Phenomena
A3 Valuing
A4 Organizing Values
A5 Internalising Values

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WEEK	DATE	[SESSION] / CHAPTER	INSTRUCTOR	NOTES
1 29 June - 3 July	Mon 29/6: 11 AM - 1 PM	[1] COURSE BRIEFING: - Syllabus overview - Course evaluation: structure of assessment components - Online assignment: Cengage WebAssign Interactive	Dr. Nor Azah	
		Ch 1: Units, Trigonometry and Vectors (Part 1) 1.1 Standards of Measurement 1.2 Measurement, Uncertainty and Estimation	Dr. Nurul Izzah	
	Tue 30/6: 2-4 PM	[2] Ch 1: Units, Trigonometry and Vectors (Part 2) 1.3 Math Review 1.4 Vectors	Dr. Nurul Izzah	
2 6-10 July	Mon 6/7: 11 AM - 1 PM	[3] Ch 2: Motion in One Dimension (Part 1) 2.1 Displacement, Velocity and Acceleration 2.2 Motion, Diagrams and Graphs	Mr. Afiq Dzuan	
	Tue 7/7: 2-4 PM	[4] Ch 2: Motion in One Dimension (Part 2) 2.3 One-Dimensional Motion with Constant Acceleration 2.4 Freely Falling Objects		
	Tutorial session	TUTORIAL CHAPTER 1	Set instructor	
3 13-17 July	Mon 13/7: 11 AM - 1 PM	[5] Ch 3: Motion in Two Dimensions (Part 1) 3.1 Displacement 3.2 Velocity, Speed and Acceleration 3.3 Two Dimensional Motion	Dr. Nurul Izzah	
	Tue 14/7: 2-4 PM	[6] Ch 3: Motion in Two Dimensions (Part 2) 3.3 Two Dimensional Motion (cont') 3.4 Relative velocity		
	Tutorial session	TUTORIAL CHAPTER 2	Set instructor	

WEEK	DATE	[SESSION] / CHAPTER	INSTRUCTOR	NOTES
4 20-24 July	Mon 20/7: 11 AM - 1 PM	[7] Ch 4: Newton's Laws of Motion (Part 1) 4.1 Force and Newton's First Law 4.2 Newton's Second Law, Gravity and Weight 4.3 Newton's Third Law	Dr. Nor Azah	TOPICAL TEST 1: Chapter 1 & 2
	Tue 21/7: 2-4 PM	[8] Ch 4: Newton's Laws of Motion (Part 2) 4.4 The Normal Force 4.5 Friction Forces 4.6 Tension Forces 4.7 Application of Newton's Laws		
	Tutorial session	TUTORIAL CHAPTER 3		
5 27-31 July	Mon 27/7: 11 AM - 1 PM	[9] Ch 4: Newton's Laws of Motion (Part 3) 4.7 Applications of Newton's Laws (Cont') 4.8 Two-Body Problems	Dr. Nor Azah	
	Tue 28/7: 2-4 PM	[10] Ch 5: Energy (Part 1) 5.1 Work 5.2 Kinetic Energy and the Work-Energy Theorem 5.3 Conservative Forces and NonConservative Forces 5.4 Gravitational Potential Energy	Dr. Nor Farhah	
	Tutorial session	TUTORIAL CHAPTER 4		
6 3-7 Aug	Mon 3/8: 11 AM - 1 PM	[11] Ch 5: Energy (Part 2) 5.5 Gravity and Nonconservative Forces 5.6 Spring Potential Energy 5.7 Systems and Energy Conservation 5.8 Power 5.8 Work Done by a Varying Force	Dr. Nor Farhah	
	Tue 4/8: 2-4 PM	[12] Ch 6: Momentum, Impulse and Collisions (Part 1) 6.1 Momentum and Impulse 6.2 Conservation of Momentum 6.3 Collisions in One Dimension	Mr. Afiq Dzuan	
	Lab session	LAB EXPERIMENT 1		
7 10-14 Aug	Mon 10/8: 11 AM - 1 PM	[13] Ch 6: Momentum, Impulse and Collisions (Part 2) 6.4 Glancing Collisions 6.5 Rocket Propulsion	Mr. Afiq Dzuan	Submission due: Lab report 1 TOPICAL TEST 2: Chapter 3 & 4
	Tue 11/8: 2-4 PM	[14] RESERVE SLOT		
	Tutorial session	TUTORIAL CHAPTER 5		

WEEK	DATE	[SESSION] / CHAPTER	INSTRUCTOR	NOTES
MID SEMESTER BREAK 17-23 AUGUST, 2026				
MID-SEMESTER EXAMINATION 24-30 AUGUST, 2026				
8	Mon 31/8: 11 AM - 1 PM	[15] 		** Public holiday 
31 Aug - 4 Sep	Tue 1/9: 9-11 AM	[16] REVISION OF CH 6	Mr. Afiq Dzuan	31 Aug. National Day
	Tutorial session	TUTORIAL CHAPTER 6	Set instructor	
9	Mon 7/9: 11 AM-1 PM	[17] Ch 7: Rotational Motion and Gravitation (Part 1) 7.1 Angular Velocity and Angular Acceleration 7.2 Rotational Motion Under Constant Angular Acceleration 7.3 Tangential Velocity, Tangential Acceleration and Centripetal Acceleration	Dr. Nor Farhah	
7-11 Sep	Tue 11/9: 9-11 AM	[18] Ch 7: Rotational Motion and Gravitation (Part 2) 7.4 Newton's Second Law for Uniform Circular Motion 7.5 Newtonian Gravitation		
	Tutorial session	TUTORIAL CHAPTER 7	Set instructor	
10	Mon 14/9: 11 AM - 1 PM	[19] Ch 8: Rotational Equilibrium and Dynamics (Part 1) 8.1 Torque 8.2 Center of Mass and Its Motion 8.3 Torque and the Two Conditions for Equilibrium 8.4 The Rotational Second Law of Motion	Dr. Nor Azah	** Public holiday 
14-18 Sep	Tue 15/9: 9-11 AM	[20] Ch 8: Rotational Equilibrium and Dynamics (Part 2) 8.4 The Rotational Second Law of Motion (Cont'); Problems Solvings		16 Sept. Malaysia Day
	Tutorial session	TUTORIAL CHAPTER 8	Set instructor	

WEEK	DATE	[SESSION] / CHAPTER	INSTRUCTOR	NOTES
11 21-25 Sep	Mon 13/10: 11 AM-1 PM	[21] Ch 8: Rotational Equilibrium and Dynamics (Part 3) 8.5 Rotational Kinetic Energy 8.6 Angular Momentum	Dr. Nor Azah	TOPICAL TEST 3: Chapter 6 & 7
	Tue 14/10: 9-11 AM	[22] Ch 9: Vibrations and Waves (Part 1) 9.1 Hooke's Law 9.2 Elastic Potential Energy 9.3 Concepts of Oscillation Rates in Simple Harmonic Motion 9.4 Position, Velocity and Acceleration as Functions of Time	Dr. Nurul Izzah	
	Tutorial session	TUTORIAL CHAPTER 8-9	Set instructor	
12 28 Sep – 2 Oct	Mon 20/10: 11 AM-1 PM	[23] Ch 9: Vibrations and Waves (Part 2) 9.5 Motion of a Pendulum 9.6 Damped Oscillations 9.7 Waves 9.8 Frequency, Amplitude and Wavelength	Dr. Nurul Izzah	
	Tue 21/10: 9-11 AM	[24] Ch 9: Vibrations and Waves (Part 3) 9.9 The Speed of Waves on Strings 9.10 Interference of Waves 9.11 Reflection of Waves	Dr. Nor Farhah	
	Lab session	LAB EXPERIMENT 2	Set instructor	
13 5-9 Oct	Mon 27/10: 11 AM-1 PM	[25] Ch 10: Sound (Part 1) 10.1 Producing a Sound Wave 10.2 Characteristics of Sound Waves 10.3 The Speed of Sound 10.4 Energy and Intensity of Sound Waves 10.5 Spherical and Plane Waves	Dr. Nor Farhah	TOPICAL TEST 4: Chapter 7 & 8 Submission due: Lab report 2
	Tue 28/10: 9-11 AM	[26] Ch 10: Sound (Part 2) 10.6 The Doppler Effect 10.7 Interference of Sound Waves 10.8 Standing Waves		
	Tutorial session	TUTORIAL CHAPTER 9-10	Set instructor	

WEEK	DATE	[SESSION] / CHAPTER	INSTRUCTOR	NOTES
14 12-16 Oct	Mon 3/11: 11 AM-1 PM	[27] Ch 10: Sound (Part 3) 10.9 Forced Vibrations and Resonance 10.10 Standing Waves in Air Columns 10.11 Beats 10.12 The Quality of Sound 10.13 The Ear	Dr. Nor Farhah	
	Tue 4/11: 9-11 AM	[28] Ch 11: Fluids and Solids (Part 1) 11.1 States of Matter 11.2 Density and Pressure 11.3 Variation of Pressure with Depth 11.4 Pressure Measurements	Dr. Nor Azah	
	Tutorial session	TUTORIAL CHAPTER 10	Set instructor	
15 19-23 Oct	Mon 10/11: 11 AM-1 PM	[29] Ch 11: Fluids and Solids (Part 2) 11.5 Buoyant Forces and Archimedes' Principle 11.6 Fluids in Motion	Dr. Nor Azah	TOPICAL TEST 5: Chapter 9 & 10
	Tue 11/11: 9-11 AM	[30] Ch 11: Fluids and Solids (Part 3) 11.7 Other Applications of Fluid Dynamics 11.8 The Deformation of Solids		
	Tutorial session	TUTORIAL CHAPTER 11	Set instructor	
16 26-30 Oct	Mon 17/11: 11 AM-1 PM	RESERVED SLOT		
	Tue 18/11: 9-11 AM	RESERVED SLOT		
	Lab session	LAB EXPERIMENT 3	Set instructor	
17 2-6 Nov	Mon 24/11: 11 AM-1 PM	RESERVED SLOT		Submission due: Lab report 3
	Tue 25/11: 9-11 AM	RESERVED SLOT		
STUDY LEAVE 9-15 NOVEMBER				
FINAL EXAMINATION 16-22 NOVEMBER 2026				